

AI Investment, EU vs US

Data, Pipeline, and the “Adopt-vs-Fund” Gap

Project documentation

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Abstract

This document explains, step by step, the empirical project behind a single claim: *Europe is becoming a consumer of artificial intelligence rather than a producer of it*. Two directly observable facts support it. First, European firms increasingly *use* AI—adoption has risen steeply since 2023. Second, almost none of the private capital that *builds* frontier AI is European: investment is overwhelmingly American, and the gap is widening. The document covers (i) the reproducible two-stage software pipeline, (ii) every data series used, with an explicit justification of its source, frequency, coverage, units, and treatment, and (iii) the descriptive charts that carry the argument. The project is *descriptive by design*: the thesis is an accounting comparison of adoption against investment, not a causal mechanism, so it runs no econometrics. Reported figures come from a clean end-to-end run; the adoption series is pulled live and may be revised by the provider, and the investment figures should be refreshed from the latest report before citation.

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1 Overview and research question

Thesis. A common framing of the EU’s position in AI fixates on the venture-capital gap with the United States. This project sharpens the point into a testable contrast: Europe *adopts* AI without *funding* it. The claim breaks into three legs, each mapped to a chart:

- (A) **Investment.** The money building AI is concentrated in the United States to an extraordinary degree, and the lead is growing, not shrinking (the headline leg).
- (B) **Adoption.** Yet European *use* of AI is rising fast—the EU is a large and growing customer for AI it largely did not build.
- (C) **Within-Europe divides.** Both the using and (what little) funding happen unevenly: a north-west/south-east geographic split, and a large-firm/SME split.

Why descriptive, not econometric. The sibling precautionary-saving project devotes most of its length to isolating a *latent* causal channel (does uncertainty *drive* saving?), because there co-movement is not causation. Here the situation is different and simpler: both halves of the thesis are *directly measured* quantities—a share of firms using AI, and a flow of dollars invested—so the argument is an accounting comparison, not an identification problem. There is no hidden variable to recover and nothing to instrument. The honest altitude is therefore a small set of clean, well-sourced charts; adding a regression would dress up a descriptive claim as a causal one. What the project owes the reader instead is rigour about *data provenance and comparability*, which Section 3 supplies.

What the documentation covers. Section 2 describes the reproducible pipeline. Section 3 justifies every data choice and states the two comparability caveats that matter. Section 4 walks through the five charts. Sections 5–6 synthesise and qualify the findings. The appendix lists the file tree, exact commands, and a data dictionary.

2 The reproducible pipeline

The project is split into two scripts by *function*, so that each stage can be run, audited, and re-run independently. Data collection (slow, network-bound, non-deterministic across vintages) is separated from figure rendering (fast, deterministic, offline). There is deliberately no third stage: per Section 1 the analysis is descriptive.

Stage / script	Reads	Writes
01_collect_data.py	Eurostat (live API) + AI Index figures	data/*.csv (tidy series)
02_make_figures.py	data/*.csv	figures/*.png (charts A–B3)

One-command reproduction. The wrapper `run_all.sh` installs the pinned dependencies and runs the two stages in order:

```
bash run_all.sh          # pip install, then run 01 -> 02 (classic Python)
```

Classic Python, no virtual environment. The project runs directly on a standard system Python (≥ 3.11); there is nothing to create or activate. `run_all.sh` installs the requirements into the chosen interpreter (python3 by default; override with the `PYTHON` variable) and then runs the two stages. Dependencies are pinned in `requirements.txt` so the charts are reproducible across machines. If the default python3 is a Homebrew / “externally-managed” build that refuses a global `pip install`, point `PYTHON` at a python.org interpreter or add `-user` to the pip line.

Network robustness. The adoption data is pulled through the keyless eurostat package. Each pull is wrapped in `try/except`: a failed source prints a clear message and leaves any CSV from a previous run untouched, so a transient outage never corrupts the rest of the dataset. The latest survey year is detected automatically from the returned table, so a new Eurostat release flows through to the country bar and the firm-size split with no code change.

3 Data

This section answers the “why this data” question. For each series we state the *source* and exact dataset code, the *frequency* and why it is appropriate, the *coverage*, the *units*, and any *transformation* and *caveat*. The adoption series is free and keyless; the investment figures are public but, as explained below, are not exposed through any free API.

3.1 Adoption: AI use by EU enterprises

Source / code

Eurostat, ISOC_EB_AI (“Artificial intelligence by size class of enterprise”), indicator E_AI_TANY (enterprises using at least one of the surveyed AI technologies), unit PC_ENT (% of enterprises), activity C10-S951_X_K (all activities excluding the financial sector). Pulled live.

Frequency / coverage

Annual, but the module is not run every year: available years are 2021, 2023, 2024, 2025 (no 2022). The latest year drives the country and firm-size charts; all years feed the trend chart.

Units

Percentage of enterprises with 10 or more persons employed (size class GE10) using ≥ 1 AI technology. This is an *extensive* measure—whether a firm uses any AI at all—not an intensity or spending measure.

Why this exact indicator

ISOC_EB_AI contains dozens of indicators (by purpose, by technology type, by barrier, by provenance). E_AI_TANY is the single headline “does the firm use any AI” switch and the one Eurostat itself reports as the adoption rate; the collector filters to it explicitly rather than blending indicators.

Caveat: series break

Eurostat broadened the list of surveyed AI technologies between the 2023 and 2024 waves. The EU-27 rate jumps from 8.1% (2023) to 13.5% (2024); part of that step is the wider definition, not pure real growth. The 2024 $\rightarrow$ 2025 step (13.5% $\rightarrow$ 19.9%) is on the consistent definition and is clean. The trend chart shades the break and the caption flags it.

3.2 Adoption by firm size (leg C)

Source

The same ISOC_EB_AI/E_AI_TANY series, geography EU27_2020, broken out by the size classes 10–49 (small), 50–249 (medium), and GE250 (large).

Why

AI adoption is sharply increasing in firm size; the small/large contrast is the cleanest within-EU statement of the “who actually uses it” question, and is robust (a single country-free EU aggregate).

3.3 Investment: private AI funding by region (legs A)

Source

Stanford HAI *AI Index* reports (2024, 2025, 2026 editions); underlying data assembled by QUID. Current-USD private investment.

Coverage / units

Three views are used: (i) the single-year cross-country bar for the latest year, 2025; (ii) the US/Europe/China anchor-year path 2023–2025; (iii) the cumulative 2013–2025 sum. All in billions of current US dollars of *private* investment.

Why hardcoded, with provenance

The underlying series is proprietary (Quid) and is *not* exposed through any free API; the public form is the figures printed in the report PDFs. The values are therefore entered as documented constants in `01_collect_data.py`, each tagged with its source figure number (e.g. AI Index 2026, Fig. 4.2.8), exactly the pattern the sibling project uses for the ECB survey constants. They are the project’s one set of “verify before publishing” numbers; refresh them from the next report edition.

Table 1: Private AI investment, US vs Europe vs China (US\$bn), the anchor years behind chart A2. “Europe” is the AI Index regional line and *includes the UK*; see the comparability note below. Each row is read from the edition shown.

year	United States	Europe	China	source edition
2023	67.2	11.0	7.8	AI Index 2024, Fig. 4.3.10 (line: EU+UK)
2024	109.1	19.4	9.3	AI Index 2025, Fig. 4.3.10 (line: Europe)
2025	285.9	20.9	12.4	AI Index 2026, Fig. 4.2.11 (line: Europe)

3.4 Two comparability caveats that matter

The two halves of the thesis are measured in *incommensurable units*—a share of firms (adoption) and a flow of dollars (investment). They are deliberately **never combined into a single ratio**; they are two parallel lenses on the same gap, presented side by side. Beyond that, two specific caveats shape how the investment numbers may be read:

- (i) **“Europe” includes the United Kingdom.** The AI Index regional time-series line bundles the UK with the EU (the 2024 edition literally labelled it “European Union and United Kingdom”), and the report publishes *no* clean EU-27 aggregate. For an EU-vs-UK split one must use the single-country bars (chart A), where the UK, France, Germany, Belgium and Sweden appear separately.

- (ii) **China is understated.** The private-investment figures exclude state “guidance funds”; an estimated US\$184 bn of such funds went to Chinese AI firms over 2000–2023. The US/China gap on *private* money therefore overstates the total-capital gap.

4 Descriptive evidence

The charts are intentionally simple; together they are the argument.

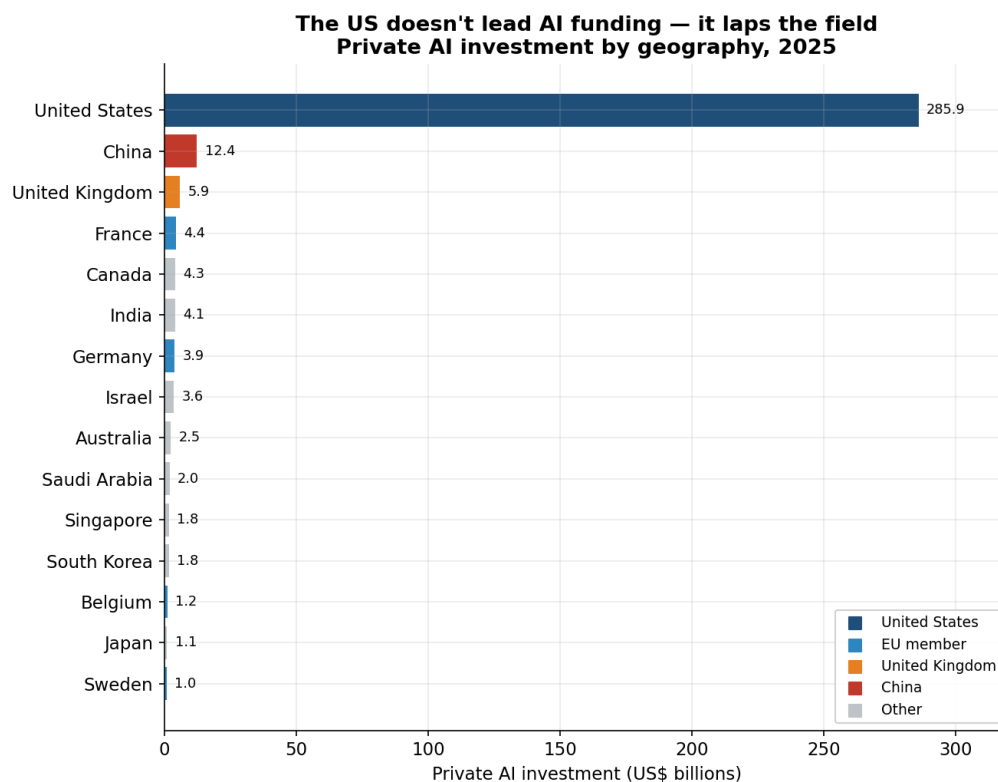


Figure 1: Leg A — the funding gap. Private AI investment by geography, 2025 (Stanford AI Index; data: Quid). The United States bar runs off the field on purpose: at US\$285.9 bn it is $\approx 23\times$ China and $48\times$ the UK. The European leaders—the UK, France, Germany, Belgium, Sweden—are slivers by comparison.

5 Interpretation

Putting the legs together:

- **Leg A (investment).** Unambiguous and stark. The United States does not merely lead private AI investment—it laps the field (Fig. 1), and the lead is *accelerating* (Fig. 2): US private AI investment more than doubled in 2025 alone, while Europe barely moved. Cumulatively since 2013 the US has attracted some US\$757 bn against Europe’s low tens of billions, and in generative AI specifically US investment ($\approx$ US\$164 bn in 2025) dwarfs all of Europe’s *total* AI investment several times over.
- **Leg B (adoption).** The mirror image. European *use* of AI is climbing fast—the EU-27 rate roughly doubled in two years to $\approx 20\%$ (Fig. 4). Europe is a large, fast-growing customer for a technology it overwhelmingly does not fund or build: a consumer, not a producer.

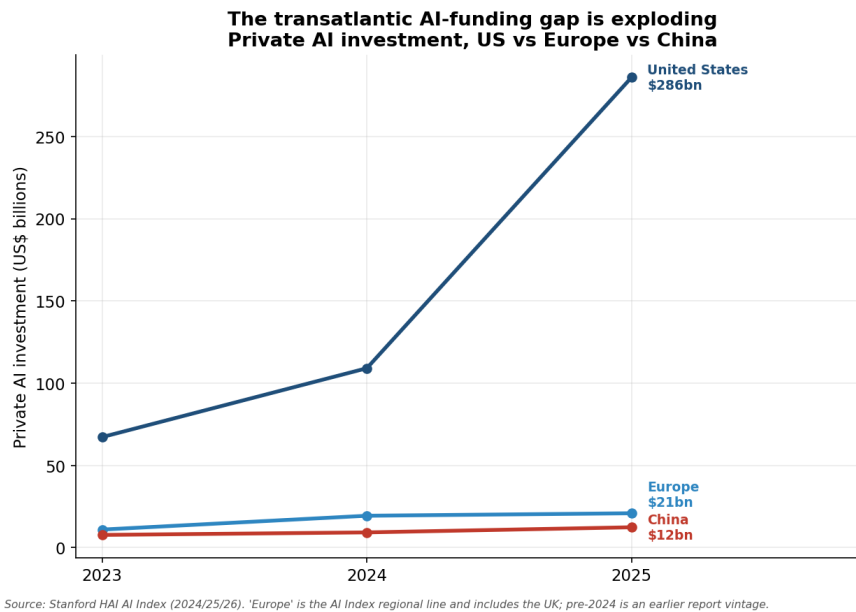


Figure 2: Leg A — the gap widening. Private AI investment, US vs Europe vs China, 2023–2025 (Table 1). The US curve goes vertical while Europe and China stay pinned near the floor; since 2024 US private investment rose $\approx 160\%$, against $\approx 7\%$ for Europe. “Europe” includes the UK, and pre-2024 is an earlier report vintage.

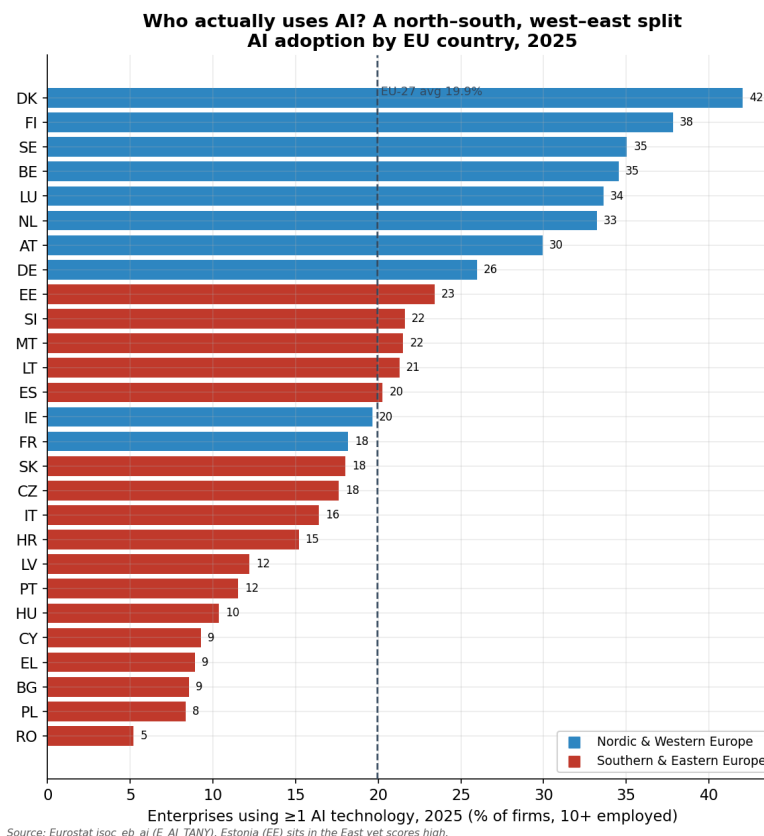


Figure 3: Leg B / C — the adoption map. Share of enterprises (10+ employed) using ≥ 1 AI technology, by EU country, 2025 (Eurostat ISOC_EB_AI). A north-west/south-east split: Denmark (42%) leads, Romania (5%) trails—an eight-fold range. The dashed line is the EU-27 average; colour marks an editorial Nordic-and-Western vs Southern-and-Eastern grouping. Estonia sits in the East yet scores high.

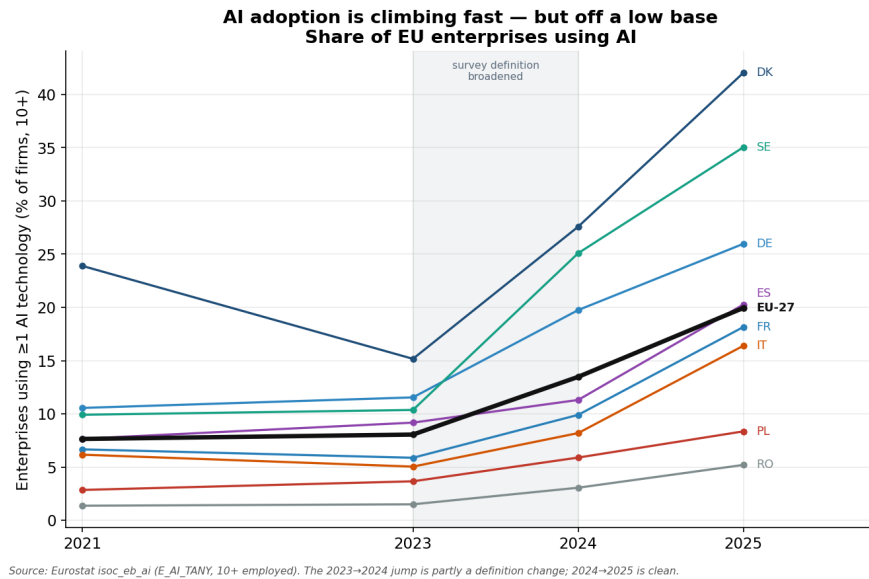


Figure 4: Leg B — rising off a low base. AI adoption over time, EU-27 (bold) plus a country spread. The EU rate roughly doubled in two years. The 2023 → 2024 segment is shaded because the survey definition was broadened then; the 2024 → 2025 rise is on the consistent definition.

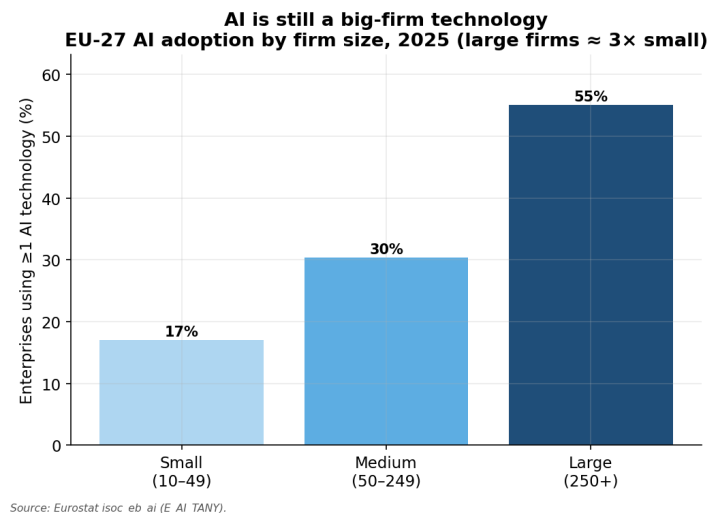


Figure 5: Leg C — the SME gap. EU-27 AI adoption by firm size, 2025. Large firms (55%) adopt at roughly three times the rate of small firms (17%): AI is, so far, a big-firm technology.

- **Leg C (within-Europe divides).** The using is uneven on two axes. Geographically, a Nordic-and-Western core (Denmark 42%) runs far ahead of the Southern and Eastern periphery (Romania 5%) (Fig. 3). By size, large firms adopt at $\sim 3\times$ the SME rate (Fig. 5). The single EU “adoption rate” hides a wide distribution.

The synthesis is the headline: *the EU is integrating AI into its economy while the economics of owning that AI accrue almost entirely elsewhere*. The adoption gap is closing; the investment gap is not.

6 Limitations and caveats

- **Series break (adoption).** The 2023 $\rightarrow$ 2024 jump partly reflects a broadened survey definition, not pure growth; lead with the clean 2024 $\rightarrow$ 2025 step or annotate the break.
- **“Europe” includes the UK (investment).** The AI Index publishes no clean EU-27 investment aggregate; the regional line bundles the UK (Sec. 3.4). EU-only figures must be built from the country bars.
- **China understated.** Private-investment figures omit state guidance funds (Sec. 3.4).
- **Extensive, not intensive, adoption.** E_AI_TANY records whether a firm uses *any* AI, not how much value it derives; a high adoption rate is not the same as deep deployment.
- **Investment data are hand-entered.** Because no free API exposes the Quid series, the investment numbers are constants read from the report PDFs and must be re-verified and refreshed each report cycle.
- **Editorial geographic grouping.** The Nordic-Western vs Southern-Eastern colouring in leg B is a simplification; the underlying continuous adoption rate is the rigorous version.
- **Vintages.** The adoption series is pulled live and revised by Eurostat; regenerate before citing any number.

A Reproduction details

A.1 File tree

```
01_collect_data.py    # STEP 1  data collection -> data/*.csv
02_make_figures.py    # STEP 2  figures          -> figures/*.png
run_all.sh            # pip install + run 01 -> 02 (classic Python, no venv)
requirements.txt       # pinned dependencies
data/                 # generated tidy CSVs
figures/              # generated charts
README.md             # short project readme
documentation.tex      # this document
```

A.2 Commands

```
# one-shot, fully reproducible
bash run_all.sh

# or stage by stage (after: pip install -r requirements.txt)
python 01_collect_data.py
```



```
python 02_make_figures.py
```

A.3 Data dictionary (generated CSVs)

file	shape	contents
ai_adoption_by_country.csv	latest year	geo, value (% firms), year
ai_adoption_trend.csv	long panel	geo, year, value (% firms)
ai_adoption_by_firm_size.csv	latest year	size_emp, value (%), year
ai_investment_by_geo.csv	2025	country, investment (US\$bn), year
ai_investment_us_eu_cn.csv	2023–25	year, US, Europe, China, source
ai_investment_cumulative.csv	2013–25 sum	country, cumulative (US\$bn)
ai_investment_genai.csv	2025	region, gen-AI investment (US\$bn)

The charts in this document come from a clean end-to-end run and are reproducible via `run_all.sh`. The adoption series is pulled live and revised by Eurostat, and the investment figures are read from periodically updated reports; regenerate the data and re-compile before citing.